

Appendix C Additional Examples

This section includes additional examples that were briefly mentioned in Section 3.

Example C.1. Cheng and Small (2006) study the special case of Example 3.1 in which $\mathcal{D} = \mathcal{Z} = \{0, 1, 2\}$. Their “Monotonicity I” assumption corresponds to Assumption 2.2 with $\mathcal{R}$ defined as in Example 3.1. They further consider imposing the restriction that $Q\{D(1) = 1 \mid D(2) = 2\} = 1$, i.e., that subjects who would comply with assignment to treatment 2 would also comply with assignment to treatment 1. They argue that this assumption is plausible in contexts where the “cost” of compliance with treatment 1 is lower than that with treatment 2. The combination of these two restrictions can be formulated in terms of Assumption 2.2 with $\mathcal{R} = \{(r_o, r_t) : (d(0), d(1), d(2)) \in \{(0, 0, 0), (0, 1, 0), (0, 1, 2)\}\}$. In their application, Cheng and Small (2006) focus on the following parameters: (i) $E_Q[Y(j) - Y(0) \mid (D(0), D(1), D(2)) = (0, 1, 2)]$ for $j \in \{1, 2\}$; and (ii) $Q\{(D(0), D(1), D(2)) = r_t\}$ for different values of r_t . Each of these parameters can be expressed in the form of (2) for appropriate choices of g and $\mathcal{R}'$. For example, the parameter in (i) equals $E_Q[g(R) \mid R \in \mathcal{R}']$ for $g(R) = Y(j) - Y(0)$ and $\mathcal{R}' = \{(r_o, r_t) : (d(0), d(1), d(2)) = (0, 1, 2)\}$. Our framework nests their parameters within their context, while also allowing for inference on the corresponding parameters for trials with more than three treatment arms, for which their approach becomes computationally infeasible. ■

Example C.2. Kirkeboen et al. (2016) study the effects of college field of study on earnings. In their setting, $\mathcal{D} = \{0, 1, 2\}$ represents three fields of study, ordered by their (soft) admission cutoffs from lowest to highest. The instrument is $Z \in \{0, 1, 2\}$, where $Z = 1$ when the student crosses the (soft) admission cutoff for field 1 but not for field 2, $Z = 2$ when the student crosses the (soft) admission cutoff for field 2, and $Z = 0$ otherwise. The authors assume that Z is exogenous in the sense that Q satisfies Assumption 2.1, and they impose:

$$Q\{D(1) = 1 \mid D(0) = 1\} = 1, \quad (159)$$

$$Q\{D(2) = 2 \mid D(0) = 2\} = 1. \quad (160)$$

These monotonicity conditions require that crossing the cutoff for field 1 or 2 weakly encourages students toward that field. They further impose the following “irrelevance” conditions:

$$Q\{D(0) = 0, D(1) = 2\} = 0 = Q\{D(0) = 2, D(1) = 0\} \quad (161)$$

$$Q\{D(0) = 0, D(2) = 1\} = 0 = Q\{D(0) = 1, D(2) = 0\}. \quad (162)$$

The condition in (161) states that crossing the cutoff for field 1 does not cause a student to switch between field 0 and field 2. A similar interpretation applies to (162). The restrictions in (159)–(162) can be formulated in terms of Assumption 2.2 by setting $\mathcal{R} = \{(r_o, r_t) :$

$(d(0), d(1), d(2)) \in \{(0, 0, 0), (0, 0, 2), (0, 1, 0), (0, 1, 2), (1, 1, 1), (1, 1, 2), (2, 1, 2), (2, 2, 2)\}$. Kirkeboen et al. (2016) acknowledge the irrelevance conditions (161)–(162) are strong. Indeed, these conditions exclude, for example, the type $(d(0), d(1), d(2)) = (0, 1, 1)$, which corresponds to a student that has a preference for field 1 over field 2 and would therefore select field 1 when above the (soft) cutoff for field 2. Nevertheless, under these conditions, Kirkeboen et al. (2016) show that the TSLS estimands identify $E_Q[Y(1) - Y(0) \mid D(1) = 1, D(0) = 0]$ and $E_Q[Y(2) - Y(0) \mid D(2) = 2, D(0) = 0]$. Under their assumptions (159)–(162), our analysis allows one to analyze additional parameters that do not correspond to the TSLS estimands, including the average relative effects of field 1 versus field 2 for any given generalized principal stratum. Our framework further allows one to investigate what can be learned about these parameters without imposing (161)–(162). For further discussion of this example and Example 3.3, see Lee and Salanié (2026) and Bai et al. (2025). ■

Example C.3 (*Revealed-Preference Restrictions*). Heckman and Pinto (2018) consider a setting in which a voucher assignment $Z \in \mathcal{Z} = \{0, 1, 2\}$ subsidizes the purchase of one of three cars, denoted by A , B , and C . They assume that the voucher assignment is randomized, so Assumption 2.1 holds. The treatment $D \in \mathcal{D} = \{A, B, C\}$ corresponds to the car purchased. In this setting, Heckman and Pinto (2018) consider a series of examples in which they apply the Weak Axiom of Revealed Preference (WARP) to the voucher-induced budget-set relationships, in the first two examples below together with an additional assumption of neutral income effects on car choice, to restrict treatment response types. Throughout the examples below, we write treatment response types in the order $(d(0), d(1), d(2))$.

- (i) In their leading example, $Z = 0$ corresponds to no voucher, $Z = 1$ corresponds to a voucher that subsidizes the purchase of car A , and $Z = 2$ corresponds to a voucher that subsidizes the purchase of either B or C . Applying WARP to the corresponding budget-set relationships, together with the additional assumption that a consumer who chooses B or C when neither is subsidized continues to choose the same car when both are subsidized, yields the seven admissible response types in Table III of Heckman and Pinto (2018). These restrictions can be formulated in terms of Assumption 2.2 with $\mathcal{R} = \mathcal{Y}^3 \times \mathcal{R}_t$, where

$$\mathcal{R}_t = \{(A, A, A), (A, A, B), (A, A, C), (B, A, B), (B, B, B), (C, A, C), (C, C, C)\}.$$

- (ii) In a second example, $Z = 0$ corresponds to no voucher, $Z = 1$ corresponds to a voucher that subsidizes the purchase of B , and $Z = 2$ corresponds to a voucher that subsidizes the purchase of B or C . The same combination of WARP, the corresponding budget-set relationships, and the neutral-income-effect assumption yields the seven admissible response types in Table V of Heckman and Pinto (2018). These restrictions can be

formulated in terms of Assumption 2.2 with $\mathcal{R} = \mathcal{Y}^3 \times \mathcal{R}_t$, where

$$\mathcal{R}_t = \{(A, A, A), (A, A, C), (A, B, B), (A, B, C), (B, B, B), (C, C, C), (C, B, C)\}.$$

- (iii) Finally, in a third example, $Z = 0$ corresponds to a voucher that subsidizes the purchase of C , $Z = 1$ corresponds to a voucher that subsidizes the purchase of B , and $Z = 2$ corresponds to a voucher that subsidizes the purchase of B or C . In this case, applying WARP to the corresponding budget-set relationships, without imposing the neutral-income-effect assumption used in the first two examples, yields the seven admissible response types in Table VI of Heckman and Pinto (2018). These restrictions can be formulated in terms of Assumption 2.2 with $\mathcal{R} = \mathcal{Y}^3 \times \mathcal{R}_t$, where

$$\mathcal{R}_t = \{(A, A, A), (A, B, B), (B, B, B), (C, A, C), (C, B, B), (C, B, C), (C, C, C)\}.$$

Heckman and Pinto (2018) provide several additional examples that fit within our framework. They establish conditions for identification of response-type probabilities and of certain counterfactual means and average treatment effects conditional on particular unions of principal strata; by contrast, our analysis allows inference on a more general class of parameters, including partially identified parameters. ■

Example C.4 (*Monotone treatment response*). Consider a setting in which the treatment admits a natural ordering and write $\mathcal{D} = \{0, \dots, K_d\}$. Suppose further that Z is randomly assigned, so Assumption 2.1 holds. Following Manski (1997), it is often reasonable to assume that $Q\{Y(j) \geq Y(k)\} = 1$ for any pair of treatments $j, k \in \mathcal{D}$ with $j \geq k$. For example, in Manski and Pepper (2000), the treatment is years of schooling and the outcome is $\log(\text{wage})$, and the restriction is that additional schooling leads to (weakly) higher wages. This restriction can be formulated in terms of Assumption 2.2 with $\mathcal{R} = \{(r_o, r_t) : y(K_d) \geq \dots \geq y(0)\}$. While Manski (1997) develops partial-identification results for unconditional treatment parameters in this context without restrictions on treatment-response types, our analysis additionally allows such restrictions to be incorporated and enables inference on parameters that condition on generalized principal strata. ■

Appendix D Details for Testable Implications

In this section, we discuss a method to obtain sharp testable restrictions of the model $\mathbf{Q}$ in terms of analytical inequalities. In this section, we assume $|\mathcal{Y}| < \infty$. We will make heavy use of the fact that a nonempty bounded polyhedron can be represented in two equivalent ways. Recall from Definition 2.1 in Bertsimas and Tsitsiklis (1997) that a polyhedron in $\mathbf{R}^k$ is a set $\{x \in \mathbf{R}^k : Ax \leq b\}$, known as the H -representation of a polyhedron. If the polyhedron is

nonempty and bounded, then Theorem 2.9 in [Bertsimas and Tsitsiklis \(1997\)](#) implies that it can equivalently be represented as the convex hull of its finite number of vertices, known as the V -representation of a polyhedron. We can compute one representation from the other using the double description method of [Avis and Fukuda \(1991\)](#), which is implemented in the `rcdd` package in R and the `mpt3` package in MATLAB.

If we represent Q by the column vector $q = (q(r) : r \in \mathcal{R})$, the model $\mathbf{Q}$ can then be represented by the polyhedron $\{q : \mathbf{1}'_{|\mathcal{R}|} q = 1, q \geq 0\}$, where $\mathbf{1}_{|\mathcal{R}|}$ denotes the $|\mathcal{R}| \times 1$ vector of ones. In the rest of this section, with some abuse of notation, we use $\mathbf{Q}$ to denote this polyhedron. Further let A_0 denote the first $|\mathcal{M}|$ rows of A . It follows from Corollary 4.1 that $\mathbf{Q}_0(P, \mathbf{Q}) \neq \emptyset$ if and only if

$$(P_{ydz} : (y, d, z) \in \mathcal{M}) \in A_0 \mathbf{Q}.$$

It follows from Corollary 2.5 in [Bertsimas and Tsitsiklis \(1997\)](#) that $A_0 \mathbf{Q}$ is a bounded polyhedron and is nonempty as long as $\mathbf{Q}$ is nonempty. In that case, Theorem 2.9 in [Bertsimas and Tsitsiklis \(1997\)](#) implies it is the convex hull of its (finite number of) vertices. Let $\text{co}(\cdot)$ denote the convex hull of a set of points. We have the following algorithm for obtaining the H -representation of $A_0 \mathbf{Q}$:

Algorithm 1 Algorithm for computing testable implications

- 1: Compute the set of all vertices of $\mathbf{Q}$, denoted by $V = \{V_i : 1 \leq i \leq n\}$.
 - 2: Compute $A_0 V = \{A_0 V_i : 1 \leq i \leq n\}$.
 - 3: Compute $\text{co}(A_0 V)$.
 - 4: Obtain the H -representation of $\text{co}(A_0 V)$.
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Step 3 in Algorithm 1 is justified by the following lemma:

Lemma D.1. *Suppose $\mathbf{Q}$ is a non-empty bounded polyhedron and V is the set of vertices of $\mathbf{Q}$. Then, $A_0 \mathbf{Q} = \text{co}(A_0 V)$.*

PROOF. We first show $\text{co}(A_0 V) \subseteq A_0 \mathbf{Q}$. Indeed, each $P \in \text{co}(A_0 V)$ could be written as $\sum_{1 \leq i \leq n} \lambda_i A_0 V_i = A_0 \sum_{1 \leq i \leq n} \lambda_i V_i$ where $\lambda_i \geq 0$ for all i and $\sum_i \lambda_i = 1$, but $\sum_{1 \leq i \leq n} \lambda_i V_i \in \mathbf{Q}$ since $\mathbf{Q}$ is a polyhedron and hence convex. To show $A_0 \mathbf{Q} \subseteq \text{co}(A_0 V)$, fix $Q \in \mathbf{Q}$. Because $\mathbf{Q} = \text{co}(V)$, $Q = \sum_{1 \leq i \leq n} \lambda_i V_i$, where $\lambda_i \geq 0$ for all i and $\sum_i \lambda_i = 1$, so $A_0 Q = \sum_{1 \leq i \leq n} \lambda_i A_0 V_i \in \text{co}(A_0 V)$. ■

Appendix E Additional Results for Sections 6 and 7

Table 8 presents the true values of $\theta(Q)$ and $\Theta_0(P, \mathbf{Q})$ in Tables 2 and 4, as well as $\theta(Q)$ and $\Theta_{0,L}(P, \mathbf{Q})$ in Table 3.

Discrete Outcomes			
Design	Parameter	θ_0	$\Theta_0(P, \mathbf{Q})$
One-sided noncompliance	$\theta_{\text{ATE}}(Q)$	0.492	[0.148, 1.030]
One-sided noncompliance	$\theta_{\text{Prob}}(Q)$	0.869	[0.620, 1.000]
Encouragement	$\theta_{\text{ATE}}(Q)$	0.085	[-1.284, 1.552]
Encouragement	$\theta_{\text{Prob}}(Q)$	0.772	[0.107, 1.000]
General-Valued Outcomes: One-Sided Noncompliance Design			
Partition	L	value	$\Theta_{0,L}(P, \mathbf{Q}_{1s})$
bin1	2	0.492	[-5.126, 7.260]
bin2	5	0.492	[-2.325, 3.463]
bin3	10	0.492	[-1.287, 2.381]
Comparison with Moment Inequalities Testing			
Design	Parameter	θ_0	$\Theta_0(P, \mathbf{Q})$
$\mathcal{Y} = \{-1, 0, 1\}$	$\theta_{\text{ATE}}(Q)$	0.189	[-0.411, 0.650]
$\mathcal{Y} = \{-1, 0, 1, 2\}$	$\theta_{\text{ATE}}(Q)$	0.0116	[-1.197, 1.195]

Table 8: True parameter values and identified sets for the simulation designs reported in Table 2, true parameter values and $\Theta_{0,L}(P, \mathbf{Q})$ for Table 3, and true parameter values and identified sets for the simulation designs in Table 4. For the general-valued outcome, the parameter is $\theta_{\text{ATE}}(Q)$.

Table 9 studies a setting with $\mathcal{D} = \mathcal{Z} = \{0, 1\}$ and either $\mathcal{Y} = \{0, 1, 2\}$ or $\mathcal{Y} = \{0, 1, 2, 3\}$. We consider the model of instrument monotonicity as in Imbens and Angrist (1994) — i.e., we set $\mathbf{Q}_{\text{mon}}$ to be the set of distributions for (R, Z) satisfying Assumptions 2.1 and 2.2 with $\mathcal{R} = \{(y(0), y(1), d(0), d(1)) : d(1) \geq d(0)\}$. The parameter of interest is the average treatment effect $\theta(Q) = E_Q[Y(1) - Y(0)]$. For each $\mathcal{Y}$, we again fix a distribution $Q \in \mathbf{Q}_{\text{mon}}$, specified in the online repository. For the design in which $\mathcal{Y} = \{0, 1, 2\}$, the confidence regions for both procedures exhibit remarkably similar average lengths and coverage properties. However, the computational cost of “FSST latent” is larger than that of “FSST ineq,” reflecting the larger dimension of A in the former procedure. For the design with $\mathcal{Y} = \{0, 1, 2, 3\}$, the “FSST ineq” confidence intervals are shorter than those based on “FSST latent” while retaining a lower computational cost.

Table 10 repeats the results in Table 6, but with the parameter λ_n in the test of Fang et al. (2023) chosen by the second recommendation in that paper, which is motivated by the law of the iterated logarithm: $\lambda_n = 1/\sqrt{\log(e \vee p) \log(e \vee \log(e \vee n))}$, where p is the number of rows of the matrix A . Across all entries, the maximum absolute difference in confidence interval endpoints is 0.028. The significance of the results at the 95% level is unchanged except for the parameter $\theta_{\text{Pr}}(tc, c|\mathcal{R}_{\text{full}})$ for long-term effects under the restriction $\mathcal{R} = \mathcal{R}_{1s, \text{nh}}$: In Table 6, the confidence interval is [0.007, 0.180], whereas in Table 10 it is [0.000, 0.184].

Table 11 considers the same parameters as in Table 6 but instead reports the p -values for

Encouragement Design with $\mathcal{Y} = \{0, 1, 2\}$								
Method	$\dim(A)$	$ \Theta_0(P, \mathbf{Q}_{\text{mon}}) $	n	95% CI length	Cov. θ_0	Cov. min Θ_0	Cov. max Θ_0	Runtime (s.d.)
FSST latent	$(12 + 2) \times 27$	0.115	3000	0.160	0.972	0.938	0.946	129.87 (30.84)
			12000	0.136	0.970	0.943	0.950	144.05 (35.56)
FSST ineq	2×2	0.115	3000	0.160	1.000	0.955	0.955	86.66 (20.77)
			12000	0.137	1.000	0.966	0.958	100.66 (25.49)

Encouragement Design with $\mathcal{Y} = \{0, 1, 2, 3\}$								
Method	$\dim(A)$	$ \Theta_0(P, \mathbf{Q}_{\text{mon}}) $	n	95% CI length	Cov. θ_0	Cov. min Θ_0	Cov. max Θ_0	Runtime (s.d.)
FSST latent	$(16 + 2) \times 48$	0.185	3000	0.330	0.994	0.978	0.978	145.74 (42.61)
			12000	0.254	0.999	0.978	0.979	162.54 (49.70)
FSST ineq	2×2	0.185	3000	0.294	0.995	0.960	0.944	96.22 (28.58)
			12000	0.240	1.000	0.958	0.953	112.45 (35.50)

Table 9: Summary results for confidence intervals under $Q \in \mathbf{Q}_{\text{mon}}$ for $\mathcal{Y} = \{0, 1, 2\}$ and $\mathcal{Y} = \{0, 1, 2, 3\}$. Each row reports the length of the identified set, average length of the 95% confidence intervals, coverage rate of the true parameter, coverage rates of the lower and upper endpoints of the identified set, and average runtime in seconds for obtaining one confidence region (with standard deviation of runtime in parentheses).

One-sided Noncompliance ($\mathcal{R} = \mathcal{R}_{1s}$)						
Strata ($\mathcal{R}'$)	Outcome	$\theta_{\text{ATE}}(tc, t \mathcal{R}')$	$\theta_{\text{ATE}}(tc, c \mathcal{R}')$	$\theta_{\text{Pr}}(tc, t \mathcal{R}')$	$\theta_{\text{Pr}}(tc, c \mathcal{R}')$	
Unrestricted ($\mathcal{R}' = \mathcal{R}$)	Short Term	[-0.124, 0.162]	[-0.101, 0.103]	[0.000, 0.242]	[0.000, 0.219]	
	Long Term	[-0.183, 0.116]	[-0.200, 0.020]	[0.000, 0.259]	[0.046, 0.296]	
Pairwise Compliers ($\mathcal{R}' = \mathcal{R}_{j,k}$)	Short Term	[-0.164, 0.205]	[-0.121, 0.047]	[0.000, 0.167]	[0.000, 0.230]	
	Long Term	[-0.201, 0.163]	[-0.239, -0.039]	[0.000, 0.200]	[0.050, 0.306]	
Full Compliers ($\mathcal{R}' = \mathcal{R}_{\text{full}}$)	Short Term	[-0.169, 0.211]	[-0.262, 0.209]	[0.000, 0.169]	[0.000, 0.258]	
	Long Term	[-0.211, 0.182]	[-0.341, 0.070]	[0.000, 0.204]	[0.000, 0.344]	

One-Sided Noncompliance + No-Harm Restriction ($\mathcal{R} = \mathcal{R}_{1s, \text{nh}}$)						
Strata ($\mathcal{R}'$)	Outcome	$\theta_{\text{ATE}}(tc, t \mathcal{R}')$	$\theta_{\text{ATE}}(tc, c \mathcal{R}')$	$\theta_{\text{Pr}}(tc, t \mathcal{R}')$	$\theta_{\text{Pr}}(tc, c \mathcal{R}')$	
Unrestricted ($\mathcal{R}' = \mathcal{R}$)	Short Term	[-0.066, 0.000]	[-0.122, 0.000]	[0.000, 0.060]	[0.000, 0.123]	
	Long Term	[-0.125, 0.000]	[-0.161, -0.027]	[0.000, 0.115]	[0.017, 0.153]	
Pairwise Compliers ($\mathcal{R}' = \mathcal{R}_{j,k}$)	Short Term	[-0.082, 0.000]	[-0.127, 0.000]	[0.000, 0.094]	[0.000, 0.127]	
	Long Term	[-0.158, 0.000]	[-0.166, -0.031]	[0.000, 0.144]	[0.027, 0.165]	
Full Compliers ($\mathcal{R}' = \mathcal{R}_{\text{full}}$)	Short Term	[-0.081, 0.000]	[-0.142, 0.000]	[0.000, 0.090]	[0.000, 0.144]	
	Long Term	[-0.150, 0.000]	[-0.194, -0.001]	[0.000, 0.152]	[0.000, 0.184]	

Table 10: 95% confidence intervals for causal parameters $\theta_{\text{ATE}}(tc, d|\mathcal{R}')$ and $\theta_{\text{Pr}}(tc, d|\mathcal{R}')$ (as defined in (30)) for different subsets of generalized principal strata $\mathcal{R}'$. Pairwise complier subset $\mathcal{R}_{j,k}$ corresponds to the (j, k) in the corresponding causal parameter — e.g., $\mathcal{R}_{j,k} = \mathcal{R}_{tc,t}$ in column 3 and $\mathcal{R}_{j,k} = \mathcal{R}_{tc,c}$ in column 4. Confidence regions were obtained through test inversion of the test in Theorem 4.2 implemented removing linearly dependent rows (as in Remark 4.1), two thousand bootstrap draws, and λ_n set to equal $1/\sqrt{\log(e \vee p) \log(e \vee \log(e \vee n))}$ in the `lpinfer` R package.

testing if $\theta_{\text{ATE}}(\cdot | \cdot) = 0$, $\theta_{\text{Pr}}(\cdot | \cdot) = 0$, and $\theta_{\text{Pr}}(\cdot | \cdot) = 1$.

One-Sided Noncompliance ($\mathcal{R} = \mathcal{R}_{1s}$)					
Strata ($\mathcal{R}'$)	Outcome	$\theta_{ATE}(tc, t \mathcal{R}')$	$\theta_{ATE}(tc, c \mathcal{R}')$	$\theta_{Pr}(tc, t \mathcal{R}')$	$\theta_{Pr}(tc, c \mathcal{R}')$
Unrestricted ($\mathcal{R}' = \mathcal{R}$)	Short Term	1.000	1.000	1.000, 0.000	0.252, 0.000
	Long Term	1.000	0.126	1.000, 0.000	0.003, 0.000
Pairwise Compliers ($\mathcal{R}' = \mathcal{R}_{j,k}$)	Short Term	1.000	0.457	1.000, 0.000	0.289, 0.000
	Long Term	1.000	0.002	1.000, 0.000	0.001, 0.000
Full Compliers ($\mathcal{R}' = \mathcal{R}_{full}$)	Short Term	1.000	1.000	1.000, 0.000	1.000, 0.000
	Long Term	1.000	0.380	1.000, 0.000	0.374, 0.000
One-sided Noncompliance + No-Harm Restriction ($\mathcal{R} = \mathcal{R}_{1s,nh}$)					
Strata ($\mathcal{R}'$)	Outcome	$\theta_{ATE}(tc, t \mathcal{R}')$	$\theta_{ATE}(tc, c \mathcal{R}')$	$\theta_{Pr}(tc, t \mathcal{R}')$	$\theta_{Pr}(tc, c \mathcal{R}')$
Unrestricted ($\mathcal{R}' = \mathcal{R}$)	Short Term	0.883	0.381	0.886, 0.000	0.413, 0.000
	Long Term	0.224	0.007	0.196, 0.000	0.005, 0.000
Pairwise Compliers ($\mathcal{R}' = \mathcal{R}_{j,k}$)	Short Term	0.851	0.471	0.848, 0.000	0.466, 0.000
	Long Term	0.133	0.002	0.137, 0.000	0.010, 0.000
Full Compliers ($\mathcal{R}' = \mathcal{R}_{full}$)	Short Term	0.852	0.603	0.858, 0.000	0.508, 0.000
	Long Term	0.140	0.049	0.120, 0.000	0.045, 0.000

Table 11: p -values for $\theta_{ATE}(tc, d|\mathcal{R}')$ and $\theta_{Pr}(tc, d|\mathcal{R}')$ (as defined in (30)) for different subsets of generalized principal strata $\mathcal{R}'$. ATE cells report the p -value for $H_0 : \theta_{ATE} = 0$. Probability cells report the p -values for $H_0 : \theta_{Pr} = 0$ and $H_0 : \theta_{Pr} = 1$, in that order, separated by a comma. Pairwise complier subset $\mathcal{R}_{j,k}$ corresponds to the (j, k) in the corresponding causal parameter — e.g., $\mathcal{R}_{j,k} = \mathcal{R}_{tc,t}$ in column 3 and $\mathcal{R}_{j,k} = \mathcal{R}_{tc,c}$ in column 4. The test is implemented after removing linearly dependent rows (as in Remark 4.1), using two thousand bootstrap draws and the default choice of penalty λ_n in the `lpinfer` R package.